STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-GM-GROTHENDIECK-ABELIAN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1252-1257 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 1259-1278, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 34 of queue_sdga.json (pb=809, stmt 1252-1257, proof 1259-1278).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 079B: sources/stacks-sdga/pinned-original-source.tex, source lines 1278–1278. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FR0: sources/stacks-sdga/pinned-original-source.tex, source lines 386–397. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a graded $\mathcal{O}$-algebra. The category $Mod(\mathcal{A})$ is a Grothendieck abelian category.*

Proof. By Lemma 0FR0 and the definition of a Grothendieck abelian category (Injectives, Definition 079B) it suffices to show that $Mod(\mathcal{A})$ has a generator. We claim that

$$\mathcal{G} = \bigoplus\nolimits_{k,U} j_{U!}\mathcal{A}_U[k]$$

is a generator where the sum is over all objects U of $\mathcal{C}$ and $k \in \mathbf{Z}$. Indeed, given a graded $\mathcal{A}$-module $\mathcal{M}$ if there are no nonzero maps from $\mathcal{G}$ to $\mathcal{M}$, then we see that for all k and U we have

$$\mathrm{Hom}_{Mod(\mathcal{A})}(j_{U!}\mathcal{A}_U[k], \mathcal{M}) = \mathrm{Hom}_{Mod(\mathcal{A}_U)}(\mathcal{A}_U[k], \mathcal{M}|_U) = \Gamma(U, \mathcal{M}^{-k})$$

is equal to zero. Hence $\mathcal{M}$ is zero. □

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.